

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Perihalan Produk:** 3-Mercaptopropionic acid  
**Product Description:** 3-Mercaptopropionic acid  
**Cat No. :** L08479  
**Synonyms** 3-Thiopropionic acid.; beta-Mercaptopropanoic acid  
**CAS No** 107-96-0  
**Molecular Formula** C3 H6 O2 S

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**

|                                        |                     |
|----------------------------------------|---------------------|
| Substances/mixtures corrosive to metal | Category 1 (H290)   |
| Acute oral toxicity                    | Category 3 (H301)   |
| Acute Inhalation Toxicity - Vapors     | Category 4 (H332)   |
| Skin Corrosion/Irritation              | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation      | Category 1 (H318)   |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H290 - May be corrosive to metals  
H301 - Toxic if swallowed  
H332 - Harmful if inhaled  
H314 - Causes severe skin burns and eye damage

### Precautionary Statements

#### Prevention

P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P390 - Absorb spillage to prevent material damage  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant liner  
P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Combustible liquid

Stench

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                | CAS No   | Weight % |
|--------------------------|----------|----------|
| 3-Mercaptopropionic acid | 107-96-0 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

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|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |

## **Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure and increased heart rate. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## **Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides.

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### **Environmental precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

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## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### **Eye Protection**

Goggles

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### **Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

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**Hygiene Measures** When using do not eat, drink or smoke Provide regular cleaning of equipment, work area and clothing

**Environmental exposure controls** Prevent product from entering drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                               |                                          |
|-----------------------------------------|-------------------------------|------------------------------------------|
| Appearance                              | Colorless, Light yellow       |                                          |
| Physical State                          | Liquid                        |                                          |
| Odor                                    | Stench                        |                                          |
| Odor Threshold                          | No data available             |                                          |
| pH                                      | No information available      | 120 g/L                                  |
| Melting Point/Range                     | 17 - 19 °C / 62.6 - 66.2 °F   |                                          |
| Softening Point                         | No data available             |                                          |
| Boiling Point/Range                     | 110 - 111 °C / 230 - 231.8 °F | @ 15 mmHg                                |
| Flash Point                             | 124 °C / 255.2 °F             | <b>Method -</b> No information available |
| Evaporation Rate                        | No information available      |                                          |
| Flammability (solid,gas)                | Not applicable                | Liquid                                   |
| Explosion Limits                        | <b>Lower</b> 1.6 Vol%         |                                          |
| Vapor Pressure                          | 0.04 mmHg @ 20 °C             |                                          |
| Vapor Density                           | No information available      | (Air = 1.0)                              |
| Specific Gravity / Density              | 1.218                         |                                          |
| Bulk Density                            | Not applicable                | Liquid                                   |
| Water Solubility                        | Soluble                       |                                          |
| Solubility in other solvents            | No information available      |                                          |
| Partition Coefficient (n-octanol/water) |                               |                                          |
| Component                               | <b>log Pow</b>                |                                          |
| 3-Mercaptopropionic acid                | -2.32                         |                                          |
| Autoignition Temperature                | 295 °C / 563 °F               |                                          |
| Decomposition Temperature               | No data available             |                                          |
| Viscosity                               | No data available             |                                          |
| Explosive Properties                    |                               | explosive air/vapour mixtures possible   |
| Oxidizing Properties                    | No information available      |                                          |
| Molecular Formula                       | C3 H6 O2 S                    |                                          |
| Molecular Weight                        | 106.14                        |                                          |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

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## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Bases. Reducing Agent. Oxidizing agent.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

The toxicological properties have not been fully investigated

#### **(a) acute toxicity;**

##### **Oral**

Category 3

##### **Dermal**

Based on available data, the classification criteria are not met

##### **Inhalation**

Category 4

| Component                | LD50 Oral               | LD50 Dermal | LC50 Inhalation            |
|--------------------------|-------------------------|-------------|----------------------------|
| 3-Mercaptopropionic acid | LD50 = 96 mg/kg ( Rat ) | -           | LC50 = 1.818 mg/L/4h (Rat) |

#### **(b) skin corrosion/irritation;**

Category 1 B

#### **(c) serious eye damage/irritation;**

Category 1

#### **(d) respiratory or skin sensitization;**

##### **Respiratory**

Based on available data, the classification criteria are not met

##### **Skin**

Based on available data, the classification criteria are not met

#### **(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

#### **(f) carcinogenicity;**

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

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(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure and increased heart rate. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component                | Freshwater Fish                          | Water Flea       | Freshwater Algae | Microtox |
|--------------------------|------------------------------------------|------------------|------------------|----------|
| 3-Mercaptopropionic acid | LC50: 88 mg/L/72h<br>(Brachydanio rerio) | EC50: 9 mg/L/48h |                  |          |

**Persistence and degradability**  
**Persistence** Readily biodegradable  
**Degradation in sewage treatment plant** Persistence is unlikely.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| 3-Mercaptopropionic acid | -2.32   | 1                             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused** Waste is classified as hazardous Dispose of in accordance with the European Directives on

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|                        |                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Products               | waste and hazardous waste Dispose of in accordance with local regulations                                                                                                                                                                                                  |
| Contaminated Packaging | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                  |
| Other Information      | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms Solutions with low pH-value must be neutralized before discharge |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| UN-No                   | UN2922                                                   |
| Hazard Class            | 8                                                        |
| Subsidiary Hazard Class | 6.1                                                      |
| Packing Group           | II                                                       |
| Proper Shipping Name    | Corrosive liquid, toxic, n.o.s. 3-Mercaptopropionic acid |

### Road and Rail Transport

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| UN-No                   | UN2922                                                   |
| Hazard Class            | 8                                                        |
| Subsidiary Hazard Class | 6.1                                                      |
| Packing Group           | II                                                       |
| Proper Shipping Name    | Corrosive liquid, toxic, n.o.s. 3-Mercaptopropionic acid |

### IATA

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| UN-No                   | UN2922                                                   |
| Hazard Class            | 8                                                        |
| Subsidiary Hazard Class | 6.1                                                      |
| Packing Group           | II                                                       |
| Proper Shipping Name    | Corrosive liquid, toxic, n.o.s. 3-Mercaptopropionic acid |

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| 3-Mercaptopropionic acid | 203-537-0 | X    | -   | X     | X    | X    | X     | X    | KE-23107 |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION



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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

25-Mar-2025

SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**